IT2011 - Artificial Intelligence and Machine Learning

Department of Information Technology, Faculty of Computing

Year 2 semester 1 (2026)

Tutorial 05

Supervised Learning Workshop 2

Decision Tree

Consider the following dataset will be used for predicting whether a customer will subscribe to a streaming service based on their Age Group, Subscription Type preference, and Device Used.

Age Group	Subscription Type	Device Used	Subscribes (Actual)	Subscribes (Predicted)
Youth	Basic	Mobile	Yes	Yes
Senior	Premium	TV	No	No
Adult	Premium	Laptop	Yes	Yes
Youth	Basic	TV	No	Yes
Adult	Basic	Mobile	Yes	Yes
Senior	Basic	TV	No	No
Youth	Premium	Laptop	Yes	Yes
Adult	Premium	TV	No	No
Senior	Premium	Mobile	No	Yes
Adult	Basic	Laptop	Yes	No

1. Construct the confusion matrix.
2. Calculate the following performance metrics:
    a. Accuracy
    b. Precision
    c. Recall
    d. F1-scoreUse the following parameter values:
3. If you are going to construct a decision tree using the **Gini Index**, calculate the **initial Gini impurity** for “Age Group”, “Subscription Type” and “Device Used” feature columns.

Use the formula:

$$\text{Gini impurity} = 1 - \sum p(i)^2$$

4. Determine which attribute should be selected as the root node using the Gini impurity criterion.

Support Vector Machine (SVM)

Consider the following dataset with six training samples for a binary classification problem

Sample	(x_1)	(x_2)	Class (y)
1	5	2	+1
2	6	3	-1
3	4	2	+1
4	8	5	+1
5	2	3	−1
6	3	4	+1

A linear SVM classifier is trained and produces the following separating hyperplane:

$$f(x) = \begin{cases} +1, & w^T x + b \geq 0 \\ -1, & w^T x + b < 0 \end{cases}$$

Assume:

$$w = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, b = -2$$

1. For each training sample, compute the decision value:
$$w^T x + b$$
2. Using the rule above, determine the predicted class label for each sample.
3. Identify which samples are misclassified (if any).

Reflection Questions

1. Why might precision be more important than accuracy in some applications?
2. What happens if the decision tree becomes too deep?
3. How does SVM handle linearly inseparable data?
4. Why are SVMs effective for high-dimensional data?
5. In what situations would you prefer Decision Trees over SVMs?